

We have seen that for an elliptical orbit,

$$r = \frac{a(1-e^2)}{1+e \cos \theta},$$

consequently, by Equation (10.25)

$$a(1-e^2) = \frac{1}{GM} \left(\frac{l}{m} \right)^2 = \frac{h^2}{\mu}.$$

$$\therefore r = \frac{h^2/\mu}{1+e \cos \theta}.$$

Measuring θ from perigee, we obtain

$$r_p = \frac{h^2/\mu}{1+e} \quad \text{and} \quad r_a = \frac{h^2/\mu}{1-e}.$$

The velocities at perigee and apogee are

$$v_p = \frac{h}{r_p} \quad \text{and} \quad v_a = \frac{h}{r_a}.$$

The energy (per unit mass) at perigee is

$$\mathcal{E} = \frac{1}{2}v_p^2 - \frac{\mu}{r_p} = \frac{\mu}{r_p} - \frac{h^2}{2r_p^2} \frac{\mu}{r_p} = -\frac{\mu}{2a},$$

where the last step is obtained from Equation (10.27).

10.10.2 The Hohmann Transfer Orbit

Satellites are usually initially placed in a circular "parking orbit," and then raised to their final orbit by means of a "transfer orbit." We now consider a simple transfer orbit called the Hohmann transfer orbit. See Figure 10.14.

If a satellite in a circular orbit of radius r_c is given an impulse by turning on a rocket motor for a short burst, its velocity will increase by Δv and the orbit is transformed into an ellipse. The perigee of the ellipse will be $r_p = r_c$. The energy will have increased to $\mathcal{E}_t$. The elliptical orbit will have a semimajor axis given by $a = -\frac{\mu}{2\mathcal{E}_t}$, and the apogee will be at $r_a = 2a - r_p$. We can

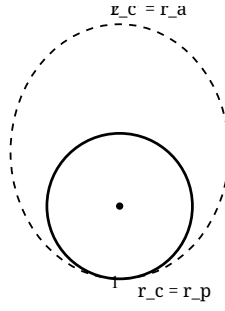


Figure 10.14 A Hohmann transfer orbit. The satellite is given an impulse at point 1 to place it in the elliptical orbit, and another impulse at 2 to place it in the circular orbit.

10.11 A PERTURBED CIRCULAR ORBIT

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assume the position of the satellite has not changed during the "burn," but now its velocity will be given by

$$v_1 = v_c + \Delta v_1 = \sqrt{2 \left(\frac{\mu}{r_p} + \mathcal{E}_t \right)},$$

where v_c is the velocity of the satellite in the initial circular orbit.

In a Hohmann transfer orbit, when the satellite reaches apogee it is given a second impulse to change the speed at apogee from $v_2 = v_a$ to that of a circular orbit of radius $r'_c = r_a$. Note that at the apogee of the elliptical transfer orbit, the speed is

$$v_2 = \sqrt{2 \left(\frac{\mu}{r_a} + \mathcal{E}_t \right)},$$

whereas the speed of a satellite in a circular orbit of radius r_a is

$$v'_c = \sqrt{2 \left(\frac{\mu}{r_a} - \frac{\mu}{2a} \right)} = \sqrt{\frac{\mu}{r_a}},$$

because $r_a = r'_c = a$. Recalling that the energy is a negative quantity, we appreciate that the satellite has to be given an additional impulse to increase its speed to that of the desired circular orbit. That is, $v'_c = v_2 + \Delta v_2$.

There are other orbit transfer schemes; the Hohmann transfer is the most economical (least amount of fuel burned) but the transfer takes a longer time than the other schemes.

Exercise 10.19

Show that $\mathcal{E} = \frac{\mu}{2a}$ implies that $r_p = a(1 - e)$. ■

Exercise 10.20

Show that the velocity of a satellite at perigee can be expressed as $v_p = \sqrt{\frac{1+e}{r_p}}$. ■

10.11 A Perturbed Circular Orbit

In this section we will study the *stability* of a circular orbit. When a physical system in an equilibrium state is perturbed by a small force, it will often oscillate about equilibrium. This is considered a *stable equilibrium*. On the other hand, if the perturbation causes the system to undergo a large change, the equilibrium is unstable. (The usual example of stable equilibrium is a marble at the bottom of a dip, and unstable equilibrium is a marble balanced on the tip of a bulge.) This section is both an application of the concepts studied in the previous sections and an introduction to the methods for dealing with small perturbations.

Consider a particle moving in a circular orbit under the action of a central force. For reasons that will eventually become clear, I will not require the force to obey the inverse square law. That is, although $\mathbf{F} = f(r)\hat{\mathbf{r}}$, the functional form of $f(r)$ will be left undetermined for now.

The equations of motion are given by generalizations of [Equations \(10.2\)](#):

$$\begin{aligned} m(\ddot{r} - r\dot{\theta}^2) &= f(r), \\ m(r\ddot{\theta} + 2\dot{r}\dot{\theta}) &= 0. \end{aligned} \tag{10.29}$$

11 Harmonic Motion

This chapter is an in-depth study of harmonic motion. You are familiar with two examples of simple harmonic oscillators, namely, a pendulum and a mass on a spring. Now you will study several different kinds of harmonic motion, including underdamped, overdamped, and critically damped oscillatory motion, as well as forced harmonic motion and the motion of coupled harmonic oscillators. The techniques you will learn here are used in nearly every branch of physics.

11.1 Springs and Pendulums

A very simple oscillatory system consists of a mass m connected to a spring of constant k . To avoid complications due to forces such as gravity or friction, assume the mass is on a perfectly frictionless horizontal surface and can slide back and forth, as shown in [Figure 11.1](#).

The kinetic energy of the mass is $T = \frac{1}{2}m\dot{x}^2$ and the potential energy of the spring is $V = \frac{1}{2}kx^2$. Therefore, as described in [Section 4.3](#) the Lagrangian for the system is

$$L = T - V = \frac{1}{2}m\dot{x}^2 - \frac{1}{2}kx^2.$$

The equation of motion is obtained from Lagrange's equation:

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{x}} - \frac{\partial L}{\partial x} = 0,$$

which yields

$$\ddot{x} + \frac{k}{m}x = 0. \quad (11.1)$$

Another simple oscillatory system is a pendulum consisting of a particle of mass m (the bob) on an inextensible, massless string of length l . See [Figure 11.2](#). The kinetic energy is

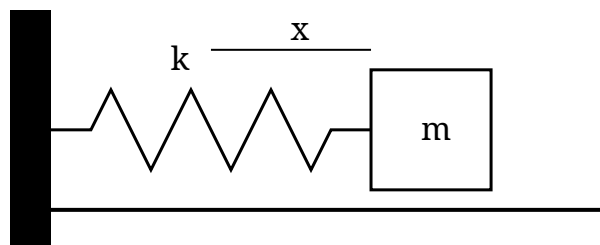


Figure 11.1 A mass on a frictionless surface connected to a spring. The distance x is measured from the equilibrium position to the center of mass of the block.

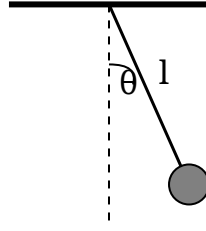


Figure 11.2 A simple pendulum.

$T = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2)$, where $x = l \sin \theta$ and $y = -l \cos \theta$ (assuming the origin is located at the point where the string is attached to the support).

$$T = \frac{1}{2}m(l^2\dot{\theta}^2 \cos^2 \theta + l^2\dot{\theta}^2 \sin^2 \theta) = \frac{1}{2}ml^2\dot{\theta}^2.$$

The potential energy is

$$V = -mgl \cos \theta.$$

Consequently, see [Equation \(4.4\)](#),

$$L = T - V = \frac{1}{2}ml^2\dot{\theta}^2 + mgl \cos \theta.$$

The Lagrange equation of motion is:

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{\theta}} - \frac{\partial L}{\partial \theta} = 0.$$

That is,

$$\frac{d}{dt}(ml^2\dot{\theta}) + mgl \sin \theta = 0,$$

or

$$ml^2\ddot{\theta} + mgl \sin \theta = 0. \quad (11.2)$$

By restricting the motion to small angles ($\theta \lesssim 20^\circ$) you can replace $\sin \theta$ by θ (in radians). Then, dividing by ml^2 ,

$$\ddot{\theta} + \frac{g}{l}\theta = 0. \quad (11.3)$$

Note that [Equations \(11.1\)](#) and [\(11.3\)](#) have exactly the same *form*; they both tell us that the second derivative of the position variable is negatively proportional to the displacement from equilibrium, meaning the system is subjected to a restoring force that is always directed toward the equilibrium point. This form for the equation of motion is characteristic of oscillatory systems. Whenever you see it, you can be sure that you are faced with a case of simple harmonic motion. You can easily show by substitution that possible solutions to the two simple harmonic motion equations above are

$$x(t) = A \cos \left(\sqrt{\frac{k}{m}}t + \beta \right) \quad (11.4)$$

and

$$\theta(t) = A \cos \left(\sqrt{\frac{g}{l}}t + \beta \right). \quad (11.5)$$

These solutions describe a sinusoidal oscillation of amplitude A and angular frequency $\omega = \sqrt{k/m}$ for the spring and angular frequency $\omega = \sqrt{g/l}$ for the pendulum. At $t = 0$ the position is given by $x(0) = A \cos \beta$. The quantity β is called the *phase constant* and is related to the initial displacement.

Consider now the somewhat more complicated case of an oscillating system subjected to a *retarding force* such as air resistance. For the sake of a specific example, let the system be a mass on a spring acted upon by a resistive force that is proportional to the velocity.¹ Then the total force acting on the mass is not just $-kx$, but rather

$$F = -kx - b\dot{x},$$

where b is a constant of proportionality. The minus sign guarantees that the retarding force opposes the motion. The $-b\dot{x}$ term is called a *damping force*. The equation of motion will now be

$$m\ddot{x} + b\dot{x} + kx = 0. \quad (11.6)$$

This is the equation of motion for a *damped harmonic oscillator* (assuming the damping force is proportional to the first power of the velocity).

Finally, a harmonic oscillator may be subjected to some external force, F_e . In that situation we simply add the external force to the other forces (but put it on the other side of the equation) and write

$$m\ddot{x} + b\dot{x} + kx = F_e. \quad (11.7)$$

This is the equation for a *forced, damped, harmonic oscillator*.

Note that these equations of motion are all linear, ordinary, second-order differential equations with constant coefficients. You learned how to solve such equations in your course on differential equations. I will give a short review of the technique, but if you are uncertain as to any of the details please review your differential equations textbook.

Exercise 11.1

(a) Prove by substitution that [Equation \(11.4\)](#) and [Equation \(11.5\)](#) are solutions to their respective equations of motion. (b) Prove by substitution that

$$x = A \cos \sqrt{k/m}t + B \sin \sqrt{k/m}t$$

is a solution to [Equation \(11.1\)](#). This is one way to express the most general solution to [Equation \(11.1\)](#). ■

Exercise 11.2

A certain spring stretches 6 cm when a force of 6 N is applied to it. A 5 kg mass is attached to the spring which is stretched 3 cm and released with a push so that its initial speed is -6 cm/s. The system lies on a frictionless horizontal surface. Find an equation for the position of the mass as a function of time. (Hint: Use the general form given in [Exercise 11.1](#).) Answer: $x = 0.030 \cos(4.47t) - 0.013 \sin(4.47t)$ m. ■

¹ Some 300 years ago, Charles Coulomb's experiments led him to conclude that kinetic friction is independent of velocity. Later more careful experiments showed that this is only approximately true. The velocity dependence of kinetic friction is still not well understood, but the frictional force for wetted surfaces is believed to be proportional to the velocity for low velocities.

Exercise 11.3

At time $t = 0$ a simple harmonic oscillator with angular frequency $\omega = 2$ rad/s is at $x = 3$ cm and has a speed of 1 cm/s. Obtain an expression for $x = x(t)$ and plot x vs. t . Answer: $x = 3 \cos 2t + (1/2) \sin 2t$ cm. ■

11.2 Solving the Differential Equation

An n th-order linear differential equation with constant coefficients has the form

$$a_n \frac{d^n x}{dt^n} + a_{n-1} \frac{d^{n-1} x}{dt^{n-1}} + \cdots + a_2 \frac{d^2 x}{dt^2} + a_1 \frac{dx}{dt} + a_0 x = f(t).$$

If the quantity $f(t)$ on the right-hand side is zero, the equation is called *homogeneous*. If $f(t) \neq 0$ the equation is called *inhomogeneous*. By "linear" we mean that the dependent variable (x) and its derivatives appear only to the first power. However, the independent variable (t) can be raised to any power. Thus $\frac{dx}{dt} = 3t^2$ is linear, but $\frac{dx}{dt} = 3x^2$ and $\left(\frac{dx}{dt}\right)^2 = 3x$ are not linear.²

11.2.1 Homogeneous Linear Differential Equations

A homogeneous differential equation of the form given above has a solution

$$x = e^{pt}.$$

The technique for obtaining the general solution is to plug $x = e^{pt}$ into the equation and carry out the indicated mathematical operations. This yields an equation for p in terms of the constants. I will illustrate this procedure in a moment. But first I want to mention that a first-order differential equation yields a single value for p , a second-order differential equation yields two values for p , and so on. Thus, depending on the order of the differential equation, you get several different solutions. Which one of these various solutions should you use? The answer is: all of them!

Rule 1. If $x_1(t)$ and $x_2(t)$ are both solutions of a linear homogeneous differential equation then $x_1(t) + x_2(t)$ is also a solution.

For simple harmonic motion we have a *second-order* differential equation so we will obtain *two* values for p . Let us call these two values p_1 and p_2 . The two solutions are:

$$x_1(t) = e^{p_1 t} \quad \text{and} \quad x_2(t) = e^{p_2 t}.$$

By Rule 1, the sum of these two solutions is also a solution. That is,

$$x(t) = e^{p_1 t} + e^{p_2 t}$$

is a solution.

We can generalize even further using another simple fact.

Rule 2. If $x(t)$ is a solution of a linear homogeneous differential equation, then $Cx(t)$, where C is an arbitrary constant, is also a solution.

Consequently,

$$x = C_1 e^{p_1 t} + C_2 e^{p_2 t}$$

is a solution to a second-order differential equation. In fact, it is the most *general* form for the solution of a homogeneous linear second-order differential equation with constant coefficients. Any other solution can be expressed as a linear combination of $e^{p_1 t}$ and $e^{p_2 t}$.

2 The "degree" of a differential equation is the power of its highest-order derivative. Thus a linear differential equation is of the first degree. The "order" of a differential equation is the order of the highest derivative appearing in the equation.

11.2 SOLVING THE DIFFERENTIAL EQUATION

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Similarly, the general solution for a third-order differential equation is

$$x = C_1 e^{p_1 t} + C_2 e^{p_2 t} + C_3 e^{p_3 t},$$

and so on.

Note that the values of p ($p_1, p_2, \dots$) come from the parameters of the problem (such as the masses and spring constants) whereas the values of the constants ($C_1, C_2, \dots$) come from the initial conditions.

Let me apply these concepts to my favorite differential equation,

$$m\ddot{x} + kx = 0.$$

If $x = e^{pt}$ then $\ddot{x} = p^2 e^{pt}$. Plug in to obtain

$$mp^2 e^{pt} + ke^{pt} = 0.$$

Dividing by e^{pt} we obtain an equation for p ,

$$mp^2 + k = 0.$$

This is called the *auxiliary equation*. Solving for p gives

$$p = \pm \sqrt{-\frac{k}{m}} = \pm i \sqrt{\frac{k}{m}} = \pm i\omega,$$

where $i \equiv \sqrt{-1}$ and $\omega = \sqrt{k/m}$.

Thus we have two (linearly independent) solutions, namely,

$$x = e^{+i\omega t} \quad \text{and} \quad x = e^{-i\omega t}.$$

Because the sum of solutions is also a solution and because we can multiply each solution by an arbitrary constant, the general solution to the second-order differential equation is

$$x = C_1 e^{+i\omega t} + C_2 e^{-i\omega t}. \tag{11.8}$$

This equation is in a perfectly acceptable form. However, it can also be expressed in terms of sines and cosines, which may be preferable in some cases.

Using "Euler's formula"

$$e^{\pm i\theta} = \cos \theta \pm i \sin \theta,$$

Equation (11.8) can be written

$$x = C_1(\cos \omega t + i \sin \omega t) + C_2(\cos \omega t - i \sin \omega t)$$

or

$$x = (C_1 + C_2) \cos \omega t + i(C_1 - C_2) \sin \omega t.$$

You might think that since x is a real quantity, you should discard the last term because it is imaginary; however, this would be unjustified because C_1 and C_2 are not yet determined and they could very well be complex numbers.

You now know basically everything you need for solving simple harmonic motion problems, except for one small detail: If the auxiliary equation for a second-order differential equation has two equal roots, then $p_1 = p_2 = p$ and the general solution is

$$x = C_1 e^{pt} + C_2 t e^{pt}. \quad (11.9)$$

This is easily verified by substitution. (Note the factor of t in the second term.)

Worked Example 11.1

Solve

$$\frac{d^2 x}{dt^2} - 4x = 0,$$

given the initial conditions $x(t = 0) = 0$ m and $\dot{x}(t = 0) = 3$ m/s.

Solution Note that this is not the SHM equation because the coefficient of x is negative. To solve, substitute e^{pt} into the equation and obtain

$$p^2 - 4 = 0.$$

Therefore, $p = \pm 2$ and the solution is

$$x = C_1 e^{2t} + C_2 e^{-2t}.$$

The initial condition $x(t = 0) = 0$ leads immediately to $C_2 = -C_1$ so

$$x = C_1 (e^{2t} - e^{-2t}) = 2C_1 \sinh(2t).$$

The initial condition $\dot{x} = 3$ leads to

$$\dot{x} = 2C_1(2) \cosh(2t) = 4C_1 \cosh(0) = 3.$$

$$\therefore C_1 = 3/4,$$

and

$$x = \frac{3}{2} \sinh(2t) \text{ m.}$$

Exercise 11.4

(a) Prove Rule 1 for a linear homogeneous second-order differential equation. (b) Prove Rule 2 for a linear homogeneous second-order differential equation. ■

Exercise 11.5

Verify that the relation $y = Ae^{ax} \cos bx + Be^{ax} \sin bx$ satisfies the equation $[(D - a)^2 + b^2]y = 0$, where $D \equiv \frac{d}{dx}$. ■

Exercise 11.6

Solve the equation

$$\frac{d^3x}{dt^3} - 4\frac{d^2x}{dt^2} + \frac{dx}{dt} + 6x = 0.$$

Answer: $x = C_1e^{-t} + C_2e^{2t} + C_3e^{3t}$. ■

Exercise 11.7

Solve $(D^2 + 2D - 3)y = 0$, where $D \equiv \frac{d}{dx}$.

Answer: $y = C_1e^x + C_2e^{-3x}$. ■

Exercise 11.8

Solve $(D^2 - 6D + 9)y = 0$, where $D \equiv \frac{d}{dt}$. ■

Exercise 11.9

Obtain formulas for $\sin 2\theta$ and $\cos 2\theta$ in terms of $\sin \theta$ and $\cos \theta$. (Hint: Note that $e^{2i\theta} = (e^{i\theta})^2$.) ■

11.2.2 An Undamped Harmonic Oscillator

Consider once again a block of mass m attached to a spring of constant k . Suppose you pull the mass until the spring is stretched to x_0 . Now release it from rest. You are asked to determine the motion, $x = x(t)$ and to evaluate the amplitude and period of the oscillations.

As given by [Equation \(11.8\)](#), the solution has the form

$$x(t) = C_1 e^{i\omega t} + C_2 e^{-i\omega t},$$

with $\omega = \sqrt{k/m}$. The only thing you need to do now is determine the constants C_1 and C_2 . The initial conditions are

$$x(t = 0) = x_0,$$

and

$$\dot{x}(t = 0) = 0.$$

Plugging these into the solution you obtain

$$x(t = 0) = x_0 = C_1 + C_2,$$

and

$$\dot{x}(t = 0) = 0 = i\omega C_1 - i\omega C_2.$$

The second equation requires that $C_2 = C_1$. Then the first equation implies that $C_1 = C_2 = \frac{1}{2}x_0$. Therefore, the solution is

$$x = \frac{1}{2}x_0(e^{+i\omega t} + e^{-i\omega t}) = \frac{1}{2}x_0(2 \cos \omega t) = x_0 \cos \omega t.$$

This equation tells us that the mass oscillates back and forth between the values $+x_0$ and $-x_0$. The quantity x_0 is called the amplitude and is usually denoted by A . The motion takes place

with a frequency

$$f = \frac{\omega}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{k}{m}}.$$

A plot of position as a function of time is presented in [Figure 11.3](#). Note that the mass returns to its original position after a time t given by

$$t = \frac{2\pi}{\omega} = \frac{1}{f} = P = \text{period}.$$

The period (P) is the time for one complete oscillation. The frequency (f) is the number of oscillations per unit time and is equal to the inverse of the period. The quantity $\omega = 2\pi f$ is called the "angular frequency" but you should be aware that often it is carelessly referred to as the frequency.

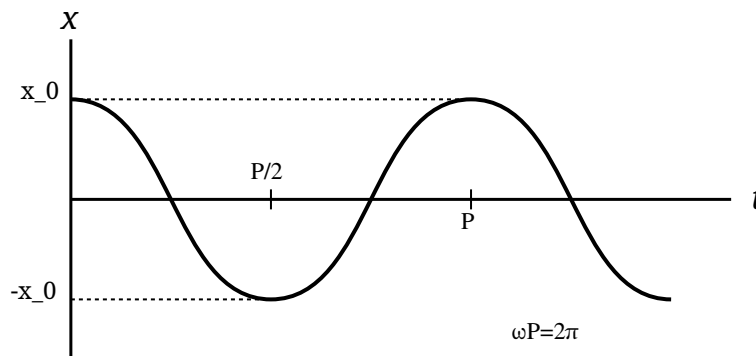


Figure 11.3 A plot of position as a function of time for a mass undergoing simple harmonic motion.

If the initial conditions are different, the form of the solution is slightly different. For example, suppose that you had been told that at $t = 0$ the block was at $x = 0$ and had a velocity v_0 . That is, $x(t = 0) = 0$ and $\dot{x}(t = 0) = v_0$. Then, using [Equation \(11.8\)](#) again you obtain

$$0 = C_1 + C_2,$$

and

$$v_0 = i\omega(C_1 - C_2).$$

Therefore,

$$C_1 = -C_2 \quad \text{and} \quad C_1 = v_0/2i\omega.$$

Consequently,

$$x = \frac{v_0}{2i\omega} (e^{i\omega t} - e^{-i\omega t}) = \frac{v_0}{2i\omega} (2i \sin \omega t) = \frac{v_0}{\omega} \sin \omega t. \quad (11.10)$$

The solution, $x = x(t)$, also gives information on the energy of the system. When $x = 0$, $\dot{x} = v_0$, and the potential energy is zero whereas the kinetic energy is a maximum. Also, when $x = x_0$ the potential energy is a maximum but the kinetic energy is zero. Since the total energy is constant,³

$$\frac{1}{2}mv_0^2 = \frac{1}{2}kx_0^2.$$

You can use this equation to relate x_0 and v_0 , thus:

$$x_0 = \pm \sqrt{\frac{m}{k}} v_0 = \pm \frac{v_0}{\omega}.$$

Therefore the result, [Equation \(11.10\)](#), can be written in the more familiar form

$$x = x_0 \sin \omega t.$$

The two examples you have just considered had very simple sets of initial conditions. In both cases C_1 and C_2 were replaced by a single constant. In general, however, the solution will contain two different constants. Then the general solution can be expressed in the form

$$x = x_0 \sin(\omega t + \beta).$$

³ Note that this result is in agreement with results from the virial theorem discussed in Worked Example 10.2. In the present problem, $n = 1$ so $\langle T \rangle = \frac{n+1}{2} \langle V \rangle$ yields $\langle T \rangle = \langle V \rangle$.